Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

Michał Pytel[1], *Paweł Pozorski*[1]

[1] Warsaw University of Technology, Poland

{michal.pytel2, pawel.pozorski}@pw.edu.pl

Abstract

Between 70–96 % of maritime casualties involve human error [1], and adverse weather compounds it by degrading both perception and the very communications crews rely on to coordinate avoidance. We present a tick-based maritime agent-based model on the real Baltic and North Sea, in which vessels follow depth- and lane-constrained shipping tracks synthesised from open bathymetry and AIS traffic-density data, and resolve encounters through a Closest-Point-of-Approach (CPA) detector coupled to a COLREGs-style give-way rule. The contribution is a quantified causal chain from *environmental and human degradation* to *collision outcome*: the success of every collision-warning exchange is gated by the product of local weather severity and a circadian crew-fatigue scalar, so tired watch-keepers in heavy weather miss warnings and collide. Collision consequences are evaluated with an accredited probabilistic damage model (DTU SIMCOL), external collision frequency is cross-validated against the regulator-grade IWRAP Mk II model, and a search-and-rescue subsystem layers IAMSAR random-search detection (MASSIM) with month-conditioned Arctic environmental degradation (Ashrafi). Against a classical and a weather-aware baseline, the system is evaluated across four scenarios with 30 seed-matched runs per condition using Mann–Whitney U tests with Holm–Bonferroni correction and Cliff's δ effect sizes. We hypothesise that fatigue-aware watch scheduling reduces the fatal-event rate by $\geq$20 % and improves post-incident survival by $\geq$15 % relative to the classical baseline.

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport carries roughly 80 % of global trade, yet the sector's safety record is dominated by a persistent human-error problem. The International Maritime Organization (IMO) attributes 70–96 % of all incidents to human causes [1], and the European Maritime Safety Agency (EMSA) identifies weather as a compounding factor in a disproportionate share of total-loss events, particularly in congested coastal corridors where crew fatigue and information latency interact [2]. Hull and machinery insurance claims exceed USD 3.5 billion annually [3]; even a 5 % reduction in fatal incidents would yield hundreds of millions in avoided liability.

The International Regulations for Preventing Collisions at Sea (COLREGs) [4] provide a deterministic collision-avoidance baseline whose implicit assumption — that crews perceive the encounter geometry accurately and communicate intentions reliably — fails under two compounding conditions that prior agent-based maritime simulation [5, 6, 7] rarely couples directly to outcome:

- **Environmental degradation of communication.** Reduced visibility, sea clutter, and heavy weather degrade the bridge-to-bridge (VHF) exchanges through which two converging vessels agree who gives way. Most collision-avoidance models assume that, once a risk is detected, the manoeuvre is always executed.
- **Human-factor coupling.** Crew fatigue accumulates over watch cycles and follows a circadian pattern with a late-night performance nadir [8]. Prior models isolate fatigue in peripheral sub-models without a direct, quantified connection to whether a warning is acted upon.

The contribution of this work is a maritime agent-based model in which the success probability of every collision-warning exchange is the *product* of an environmental factor and a crew-fatigue factor, creating a quantified causal link from *information quality under stress* to *collision and survival outcomes*. We build the model on the real Baltic and North Sea geography (Section 3), drive vessels along hand-authored shipping tracks, and resolve encounters with a CPA/TCPA detector and a COLREGs-style give-way rule (Section 5). Collision consequences use the accredited DTU SIMCOL probabilistic damage model (Section 6); the resulting collision frequency is cross-validated against IWRAP Mk II (Section 7); and a search-and-rescue subsystem (Section 8) closes the loop from incident to survival. The engine is implemented in Rust on the krABMaga agent-based framework [9] for deterministic, data-parallel batch execution. Figure 1 summarises the causal pipeline that ties these components together.

2. Research Objectives

The central thesis is: *coupling crew-fatigue dynamics to the reliability of collision-warning communication, and managing fatigue through smart watch scheduling, reduces fatal maritime outcomes more effectively than weather-unaware classical practice.*

Experimental conditions. All experiments compare three methods, implemented as configuration switches over a single agent codebase:

Baseline A (Classical) represents current weather-unaware practice: vessels follow COLREGs give-way with nominal fair-weather VHF, no weather-induced slowdown, and an unmanaged fatigue build-up. Heavy-weather comms degradation is environmental (Section 5) and therefore still applies to Baseline A inside the storm corridor — it has no privileged radio; what it lacks is the *behavioural* mitigation (speed reduction, managed

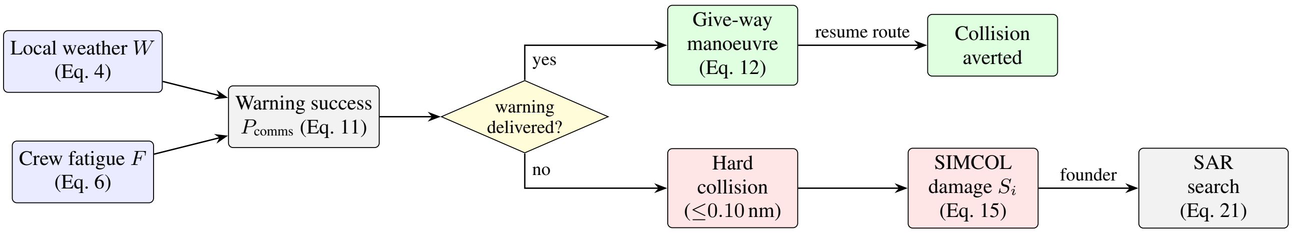

Figure 1: *Causal pipeline of the model. Local weather W and crew fatigue F jointly set the collision-warning success probability P_{comms} (Eq. 11); a delivered warning yields a give-way manoeuvre, a lost warning leaves the encounter to geometry and may produce a hard collision. Collision consequences are evaluated by the SIMCOL damage model, and a foundering vessel enters the search-and-rescue chain.*

fatigue) the other methods apply.

Baseline B (Shore Rest Alerts) adds weather-aware speed reduction and shore-issued rest advisories that slow fatigue accumulation, but no further mitigation.

Proposed System adds smart watch scheduling (lowest fatigue build-rate) on top of the weather-aware mitigations of Baseline B. Performance is measured using six KPIs defined in Table 3.

Falsifiable hypotheses. They drive all design decisions; any component that contributes to none of them is excluded.

H1. Fatigue-aware scheduling (Proposed) reduces the fatal-event rate (Table 3, KPI 1) by $\geq$20 % versus Baseline A ($p < 0.05$, Mann–Whitney U, Cliff's $|\delta| \geq 0.2$).

H2. Earlier, better-coordinated avoidance and a functioning rescue chain improve post-incident survival (Table 3, KPI 4) by $\geq$15 % versus Baseline A.

H3. The Proposed System outperforms Baseline B on the fatal-event rate ($p < 0.05$) under the storm-corridor scenario, where comms degradation is most severe.

3. System Model

3.1. World Geometry and Projection

The simulation occupies the real Baltic and North Sea basin, bounded by latitude $\phi \in [50.5°, 66.0°]$ N and longitude $\lambda \in [-5.0°, 31.0°]$ E — from the English Channel and Dutch coast in the south-west to the Gulf of Finland in the east and the mid-Norwegian coast in the north. The domain is mapped with an equirectangular projection carrying a cosine-of-mid-latitude correction, so that one field unit equals exactly one nautical mile on both axes at the reference parallel $\phi_0 = \frac{1}{2}(\phi_{\min} + \phi_{\max}) = 58.25°$. With $k_{\text{lat}} = 60$ nm/° and $k_{\text{lon}} = 60 \cos \phi_0$, a geodetic point projects to field coordinates

$$x = (\lambda - \lambda_{\min})\, k_{\text{lon}}, \qquad y = (\phi - \phi_{\min})\, k_{\text{lat}}. \tag{1}$$

The domain measures 930 nm north–south (15.5° × 60) and ≈1136 nm east–west; the residual east–west distortion is at most ∼6–10 % at the box edges and is accepted without correction terms, so the engine treats all distances as Euclidean nautical miles. The coastline is loaded from Natural-Earth polygons into an R-tree spatial index supporting accelerated point-in-polygon (land) and segment-intersection (grounding) tests. Sea depth, used only offline for route planning (Section 3.4), is taken from a separate gridded bathymetry product and is not consulted during the tick loop.

3.2. Time Resolution

Time advances in discrete ticks of $\Delta t = 15$ min (0.25 h). The default horizon is $T = 2880$ ticks (30 days). A wall-clock-of-day variable, initialised at 06:00 UTC, advances 0.25 h per tick and drives the circadian fatigue term (Section 4). All stochastic draws use a single per-run seeded generator (`SmallRng`), so identical seeds reproduce byte-identical KPI logs.

3.3. Agent Types and Voyage Logic

The model contains vessel agents and, on incident, rescue agents (Section 8). Each vessel i carries position $\mathbf{p}_i$, an ordered route $R_i = (\mathbf{w}_0, \ldots, \mathbf{w}_{m-1})$ of waypoints, a route index and travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, a discrete state $\sigma_i \in \{\textsc{Active}, \textsc{Docked}\}$, crew count $n_i = 10$, and an internal fatigue scalar $F_i \in [0, 1]$.

Routes are *synthesised procedurally* rather than replayed from raw AIS or drawn by hand: a deterministic, depth- and lane-aware planner (Section 3.4) generates ordered key-point polylines stored as `data/ais_paths.json` with schema `mmsi/name/vessel_type/waypoints[]`. This keeps the traffic substrate controlled and reproducible for the information-quality experiment, while guaranteeing that every lane is deep enough for its vessel class and follows the corridors real traffic uses. Ports are a generated set of polygons (`data/ports.json`) placed on real harbours; a vessel that enters a port polygon docks.

Cruise speed is drawn at spawn by vessel type ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases} \tag{2}$$

Waypoint pursuit. With active target $\mathbf{t}$ (an avoidance waypoint if queued, else the current route waypoint), $\boldsymbol{\delta} = \mathbf{t} - \mathbf{p}_i$ and $\rho = \|\boldsymbol{\delta}\|$, a waypoint is reached when $\rho < 0.5$ nm. The per-tick step is $s = \min(v_i\, \Delta t,\ \rho)$ and the proposed move is $\mathbf{p}'_i = \mathbf{p}_i + (\boldsymbol{\delta}/\rho)\, s$, with compass heading $\psi_i = (90° - \operatorname{atan2}(\delta_y, \delta_x)) \bmod 360°$. To prevent tunnelling through thin landmasses, the move is accepted at the first of up to five geometrically halved step lengths $s \cdot 2^{-j}$ ($j = 0, \ldots, 4$) whose segment does not cross land; a post-move grounding guard reverts $\mathbf{p}_i$ to the last valid position if it ended on land. On reaching a port or route endpoint a vessel docks for a randomised dwell $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses ($d_i \leftarrow -d_i$). A fixed fleet size N is maintained: when $|\text{fleet}| < N$, a fresh vessel is spawned onto a random hand-authored track.

Same-destination following. Co-directional traffic bound for the same destination is not separated by the give-way rule (an overtaking pair has near-zero relative velocity, Section 5), so a leader–follower discipline prevents vessels from drawing up side-by-side and overlapping. Two vessels share a destination when their route endpoints (in the travel direction) lie within 5 nm of each other. A trailing vessel within a vicinity $r_v = 3$ nm of such a leader — the vessel ahead, i.e. closer to the shared destination — caps its speed to the leader's, so it matches speed and does not overtake; if it closes within a keep-distance gap $g = 0.5$ nm it holds station (anchors). On first acquiring a leader it is radioed to slow down, match speed, and keep distance. This applies anywhere the courses converge, not only at the port mouth, and yields a stop-and-go queue whose throughput matches the leader's speed.

3.4. Route Generation from Bathymetry and Traffic Density

Rather than replaying raw AIS or hand-drawing lanes, the traffic substrate is *synthesised* by a deterministic, depth-aware route planner, so the geometry is reproducible yet physically defensible for large vessels. Two open datasets ground it. Sea depth comes from the EMODnet Digital Bathymetry (DTM), a gridded product at $1/16'$ ($\approx$115 m) resolution covering the whole basin [10]. Empirical shipping lanes come from the EMODnet Human Activities vessel-density maps — AIS-derived ship-hours per km^2 resolved by vessel type [11], consistent with the use of AIS density to characterise traffic patterns [6]. Both are fetched once through their open OGC Web Coverage Services (`ows.emodnet-bathymetry.eu` and `ows.emodnet-humanactivities.eu`) and bundled as compact grids by `tools/fetch_bathymetry.py` and `tools/fetch_lanes.py`; generation itself runs offline. Ports are a built-in gazetteer of 42 major Baltic / North-Sea harbours, each snapped onto navigable water.

Navigability by draft. Each vessel class c has a static draft T_c and a required under-keel clearance UKC$_c$, following harbour-approach-channel practice [12]. A grid cell of charted depth h is navigable for the class iff $h \geq T_c + \mathrm{UKC}_c + m_s$, with a coarse-grid safety buffer $m_s = 2$ m; Table 1 gives the three classes (deep-draft VLCCs are excluded, as they cannot transit the Danish Straits into the Baltic). Because navigability is a *depth* test, shallow archipelago water is impassable regardless of whether individual skerries are resolved in the coastline polygons — structurally preventing the routes from cutting across islands or shoals.

Table 1: *Vessel-class minimum transit depth (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC$_c$ (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

Cost-field planner. Between a sampled origin/destination port pair the planner runs a weighted A* search [13] on a 1 nm grid. Each navigable cell carries a cost multiplier

$$\mu = 1 + w_h\, s_h + w_d\, s_d + w_\ell\, (1 - a_\ell), \tag{3}$$

penalising cells whose depth is close to the class limit ($s_h \in [0, 1]$, a squat / safety margin), cells within a 2.5 nm offing of shore or shoal ($s_d \in [0, 1]$, from a multi-source distance transform of the impassable cells), and cells *off* the class's lane ($a_\ell \in [0, 1]$ the normalised vessel-density attractiveness for that class). The cost of a move is the inter-cell distance times the mean of the two endpoint multipliers; since $\mu \geq 1$ the octile distance remains an admissible heuristic. With weights $(w_h, w_d, w_\ell) = (3, 4, 2.5)$ the least-cost paths hold deep water down the middle of a channel, keep an offing from the coast, and gravitate to the corridors that class of ship actually uses.

Post-processing and determinism. Each cell path is simplified (Ramer–Douglas–Peucker) under the constraint that no shortcut segment crosses water shallower than the class allows, and a final check against the raw bathymetry (sampled every 0.15 nm) guarantees the emitted polyline never dips below T_c + UKC$_c$. A small seeded per-route cost jitter makes the output deterministic for a given seed yet varied between seeds. On the default configuration the planner emits 80 routes (35 cargo, 25 passenger, 20 tanker); verification confirms 0 of $\sim 10^5$ along-route samples below the class depth requirement, an $\approx$84 % higher mean on-lane fraction than a depth-only planner, and correct transits of the deep-water Great Belt and Skagerrak between the North Sea and the Baltic.

3.5. Weather and Hazard Field

A 50×50-cell grid evolves three physical channels — sea state s, visibility v, and wind u, each in $[0, 1]$ — which are fused into a scalar hazard $W_{ij} \in [0, 1]$:

$$W = \mathrm{clip}\left(\frac{w_s\, s + w_v\, (1 - v) + w_u\, u}{w_s + w_v + w_u},\, 0,\, 1\right), \tag{4}$$

with weights $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. A vessel samples W at its current cell only (nearest-cell lookup), so each vessel acts on local conditions. The field advances each tick through seven ordered stages (full parameters in Section 10.4): (i) a two-state calm/storm regime Markov chain selecting mean-reversion targets; (ii) up to eight Lagrangian storm cells with growth, decay, drift and removal; (iii) an AR(1)-with-advection background per channel; (iv) a system overlay imprinting storm cells onto the channels; (v) cross-channel coupling; (vi) 5-point smoothing with a gradient limiter; and (vii) hazard fusion via Eq. 4.

Independently of the stochastic field, the **storm corridor** scenario carries a single moving storm zone of radius r_{storm} that drifts at 0.30 nm/tick along a fixed lat/lon track (English Channel $\to$ North Sea $\to$ Skagerrak $\to$ Kattegat $\to$ Baltic $\to$ Gdańsk). Inside the zone, both communication reliability (Section 5) and vessel speed (Section 10.3) are degraded.

4. Crew Fatigue Dynamics

Each vessel carries an internal fatigue scalar $F_i \in [0, 1]$ (0 rested, 1 exhausted), updated every tick. The circadian drive peaks near 03:00 and troughs near 15:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\Big(1 - \cos\big(\tfrac{2\pi(t_{\text{utc}} - 3)}{24}\big)\Big) \in [0, 1]. \tag{5}$$

While ACTIVE, with local hazard W, fatigue accumulates as

$$\Delta F = \big(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\big)\, m(\text{method}), \tag{6}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015, 0.0040, 0.0008)$ and a method build-rate

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A (unmanaged)} \\ 0.82 & \text{Baseline B (shore rest alerts)} \\ 0.65 & \text{Proposed (smart scheduling)}, \end{cases} \quad (7)$$

and $F_i \leftarrow \min(1,\ F_i + \Delta F)$. While DOCKED, the crew recovers at $F_i \leftarrow \max(0,\ F_i - 0.025)$ per tick (full recovery in ≈40 ticks). The method factor m is the sole lever distinguishing the three conditions' human-factor behaviour, isolating the effect of fatigue management.

Fatigue → communication degradation. Tired watchkeepers miss or delay warnings. With threshold $F^* = 0.40$ and maximum penalty $\kappa = 0.65$,

$$g(F_i) = \begin{cases} 1, & F_i \le F^* \\ 1 - \kappa \dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \quad (8)$$

This factor multiplies into the collision-warning success probability (Eq. 11), the mechanism through which fatigue translates into collisions.

5. Collision Avoidance

5.1. CPA / TCPA Detection

Each ACTIVE pair is screened with the standard linear-motion closest-point-of-approach result. The field-frame velocity for heading ψ and speed v is $\mathbf{v}(\psi, v) = v\,\Delta t\,(\sin\psi,\ \cos\psi)$. For a pair (a, b) with relative position $\Delta\mathbf{p} = \mathbf{p}_b - \mathbf{p}_a$ and relative velocity $\Delta\mathbf{v} = \mathbf{v}_b - \mathbf{v}_a$,

$$t^* = -\frac{\Delta\mathbf{p} \cdot \Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \|\Delta\mathbf{p} + t^*\,\Delta\mathbf{v}\|, \quad (9)$$

with t^* measured in ticks. Degenerate cases return $(d_{\text{CPA}}, t^*) = (\|\Delta\mathbf{p}\|, \infty)$ when $\|\Delta\mathbf{v}\|^2 < 10^{-9}$ (parallel) and $(\|\Delta\mathbf{p}\|, 0)$ when $t^* \le 0$ (diverging). A manoeuvre is considered iff *all* of

$$\|\Delta\mathbf{p}\| \le 20\ \text{nm}, \quad d_{\text{CPA}} \le 5\ \text{nm}, \quad 0 < t^* \le 12\ \text{ticks}, \quad (10)$$

hold and the give-way vessel is not in avoidance cooldown.

5.2. Give-Way Rule and Communication Success

The vessel with the lower identifier is the give-way (turn-to-starboard) vessel; the other stands on. A deterministic lower-id rule is a reproducible surrogate for the COLREGs crossing/head-on stand-on/give-way assignment [4, 5]. Crucially, the manoeuvre is executed *only if the warning is successfully communicated*, with success probability the product of an environmental factor and a fatigue factor over both vessels:

$$P_{\text{comms}} = \underbrace{\min\big(q(\mathbf{p}_a),\ q(\mathbf{p}_b)\big)}_{\text{weather / storm}} \cdot \underbrace{\min\big(g(F_a),\ g(F_b)\big)}_{\text{fatigue, Eq. 8}}, \quad (11)$$

where $q(\mathbf{p})$ is the nominal link reliability, reduced to the storm-comms rate (as low as 0.08 inside the storm corridor — 92 % of warnings lost) when $\mathbf{p}$ lies in the storm zone. The manoeuvre proceeds iff a uniform draw is below P_{comms}; otherwise the give-way vessel fails to act, leaving the encounter to resolve by geometry alone. Equation 11 is the model's central coupling: it makes warning delivery jointly contingent on weather and human fatigue.

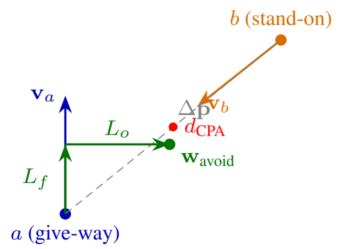

Figure 2: *Give-way geometry. The lower-id vessel a turns to starboard toward a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard of its heading (Eq. 12), opening the closest-point-of-approach distance d_{CPA} to the stand-on vessel b. The manoeuvre executes only if the warning is delivered (Eq. 11).*

5.3. Avoidance Geometry and Message Exchange

A give-way vessel receives one temporary waypoint placed $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of its heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \quad (12)$$

A three-message exchange is logged — `CollisionWarning{cpa,tta}` → `GivingWay` → `MaintainingCourse`, with `ResumeRoute` once the avoidance waypoint is consumed. Each issued warning records its t^* for the average-TTA KPI, and a 3-tick cooldown prevents manoeuvre thrashing.

5.4. Optional Continuous-Avoidance Track (Force Field)

As an optional realism layer, the discrete give-way waypoint of Eq. 12 can be replaced by a continuous artificial-force-field manoeuvre in the style of Xiao et al. [14], in which COLREGs behaviour emerges rather than being scripted. The track is a default-off configuration switch; when enabled it supersedes the CPA waypoint avoidance for that run, leaving every other subsystem unchanged. A vessel responds to a repulsive force from each nearby obstacle (vessel or shoal) within an effect distance d_o,

$$\mathbf{F}_{\text{rep}} = \sum_{k:\,d_k < d_o} \frac{k_{\text{rep}}}{d_k^p}\,\hat{\mathbf{n}}_k, \quad (13)$$

where d_k is the distance to obstacle k, $\hat{\mathbf{n}}_k$ the unit vector away from it, k_{rep} a steepness constant, and p the repulsion exponent. The resultant force's lateral (starboard) component sets a rudder angle that drives a first-order Nomoto manoeuvring response with non-dimensional indices $K' = K(L/v)$, $T' = T(v/L)$ [14]; a small starboard bias resolves the symmetric dead-ahead head-on to the COLREGs "both alter to starboard" outcome. The crew-preparedness state scales the commanded rudder, so a fatigued crew yields weaker, later avoidance. The exponent p is exposed as a tunable parameter defaulting to the inverse-square form ($p = 2$): the source sets the parameters of Eq. 13 by coarse optimisation and states their AIS calibration is still in development, so no single value is pinned by it. Xiao's calibrated effect distances are micro-scale—$d_o \sim \mathcal{N}(1548,\ 706^2)$ m for head-on and $\mathcal{N}(384,\ 358^2)$ m for overtaking encounters—for a sub-10 km waterway stepped each second; at this model's basin scale and 15-minute tick they fall below a single step, so the magnitudes are rescaled to operationally useful reaches while preserving the source's ≈ 4:1 head-on-to-overtaking ratio.

6. Collision Consequence Model (SIMCOL)

A **hard collision** is logged when two ACTIVE vessels close within $d_{\text{hit}} = 2\, r_{\text{coll}} = 0.10\,\text{nm}$ ($\approx$185 m, roughly a ship-length sum), counted once per encounter — on the tick the pair *enters* contact, not every tick it remains overlapping. Encounters within 1 nm of any port are treated as VTS-controlled harbour manoeuvring and are not counted as collisions. The consequence of a hard collision is evaluated with the DTU SIMCOL probabilistic damage and survivability model [15], which replaces an ad-hoc casualty rule with collision mechanics.

External dynamics. The energy released for hull deformation is the loss of kinetic energy of the rigid-body system between first contact and the common-velocity state, evaluated with the closed-form external-dynamics solution of Pedersen and Zhang [16] as used by Lützen [15, 17]. Only the component of the closing velocity *normal* to the struck vessel's side performs side-penetrating work, so for struck and striking displacements M_a, M_b the absorbed energy is the reduced-mass energy on that normal component,

$$E_{\text{abs}} = \tfrac{1}{2}\,\frac{M'_a\,M'_b}{M'_a + M'_b}\,V_n^2, \qquad V_n = \big|(\mathbf{v}_b - \mathbf{v}_a)\cdot\hat{\mathbf{n}}\big|, \quad (14)$$

where $\hat{\mathbf{n}}$ is the unit normal to the struck side and $M'_a = M_a(1 + m_{\text{sway}})$, $M'_b = M_b(1 + m_{\text{surge}})$ carry hydrodynamic added mass with sway coefficient $m_{\text{sway}} = 0.85$ and surge coefficient $m_{\text{surge}} = 0.05$ [15]. The normal-velocity projection V_n supplies the collision-angle dependence directly — a near-parallel graze releases almost no energy while a perpendicular impact releases the maximum — so Eq. 14 reduces to the classical right-angle reduced-mass form when $\hat{\mathbf{n}}$ aligns with the closing velocity, and no separate oblique-reduction factor is required.

Internal mechanics. The absorbed energy is related to damage extent through Minorsky's absorbed-energy/resistance-volume correlation [18], $E_{\text{abs}}\,[\text{MJ}] = 47.2\,R_T + 32.7$, where R_T is the resistance (destroyed) volume of deformed structural material. Lützen casts the resulting damage as *non-dimensional* distributions — the damage length ℓ/L_{pp} (longitudinal extent normalised by struck-ship length) and the penetration t/B (transverse extent normalised by breadth), each fitted to striking-ship displacement, speed, and collision angle [15]. Taking a representative damage length $\ell = 0.12\,L_{pp}$ (the median of that distribution) and an effective smeared side-steel thickness $\tau = 0.08\,\text{m}$, the breach penetration is $t = R_T/(\ell\,\tau)$, expressed non-dimensionally as t/B.

Survivability. Because the engine does not carry per-ship residual-stability curves, the survivability factor $S_i \in [0, 1]$ — the conditional probability that the struck vessel survives the flooding case — is modelled as a SOLAS II-1-shaped surrogate [19] on the relative penetration:

$$S_i = \begin{cases} 1, & t/B \le r_0 \\ \left(\dfrac{r_1 - t/B}{r_1 - r_0}\right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \ge r_1, \end{cases} \quad (15)$$

with knee $r_0 = 0.10$, full-loss penetration $r_1 = 0.30$, and the 0.25 exponent of the SOLAS II-1 s-factor. The expected number of crew thrown into the water (man-overboard) per collision is then

$$\mathbb{E}[\text{overboard}] = n_b\,\big(1 - S_i\big), \quad (16)$$

with n_b the struck-ship crew. Each crew member is an independent Bernoulli trial with overboard probability $1 - S_i$, and each colliding vessel is evaluated independently as the struck ship. This draw does *not* kill directly: it determines how many crew enter the water, whose fate is decided by the search-and-rescue chain (Section 8). The consequence then follows one of two parallel casualty paths. A vessel whose survival factor falls below the founder threshold $S^\dagger = 0.50$ founders, evacuating its *whole* crew to a liferaft (the EVAC chain); a struck vessel that stays afloat instead casts only its $\mathbb{E}[\text{overboard}]$ casualties into the water as individual persons-in-water for the man-overboard search (Section 8.3). A fatality therefore arises only when a person in the water is not recovered within the survival window; $S_i \to 1$ is logged as a near-miss damage event.

Encounter geometry. Every logged collision is additionally classified by its encounter geometry, taken from the angle between the two vessels' headings folded onto $[0°, 180°]$, $\theta = \min(|\psi_a - \psi_b|,\, 360° - |\psi_a - \psi_b|)$:

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45° \\ \text{front-to-side (crossing)}, & 45° \le \theta < 135° \\ \text{head-on}, & \theta \ge 135°. \end{cases} \quad (17)$$

Per-type counts are accumulated as diagnostic KPIs (Section 9). The classification is consistent with the consequence chain: near-parallel side-to-side hits have $\hat{\mathbf{n}}$ nearly orthogonal to the closing velocity, so $V_n \to 0$ in Eq. 14 and they release little energy, whereas head-on and crossing encounters carry the large normal closing speed that drives deep penetration and foundering.

7. External Collision-Frequency Validation (IWRAP)

The internal KPIs (Section 9) measure relative differences between methods but are not, by themselves, regulator-grade frequencies. To anchor the model to accepted practice, each run's vessel tracks are exported as synthetic AIS and analysed offline with the IWRAP Mk II model [20]. The expected number of collisions on a waterway is the product of a geometric candidate count and a causation probability [21, 20]:

$$N_c = N_a\,P_c. \quad (18)$$

For two opposing flows of classes i, j over a leg of sailing distance D and fairway width W during time t, the head-on candidate count is

$$N_a^{\text{head}} = \frac{D\,t}{W}\sum_{i,j} Q_i^{(1)} Q_j^{(2)}\,\frac{(B_i^{(1)} + B_j^{(2)})\,(V_i^{(1)} + V_j^{(2)})}{V_i^{(1)}\,V_j^{(2)}}, \quad (19)$$

with traffic flows Q, beams B, speeds V; the overtaking form replaces $(V_i + V_j)$ with $|V_i - V_j|$. For crossing waterways meeting at angle $\Theta \in (10°, 170°)$,

$$N_a^{\text{cross}} = \sum_{i,j} Q_i^{(1)} Q_j^{(2)}\,\frac{D_{ij}\,V_{ij}}{V_i^{(1)}\,V_j^{(2)}}\,\frac{1}{\sin\Theta}, \quad (20)$$

where $V_{ij} = \sqrt{(V_i^{(1)})^2 + (V_j^{(2)})^2 - 2V_i^{(1)}V_j^{(2)}\cos\Theta}$ is the relative speed and D_{ij} the Pedersen collision diameter [20, 17]. The causation probabilities are the IWRAP software defaults [20, 21]: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, and $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$. An offline analyzer builds

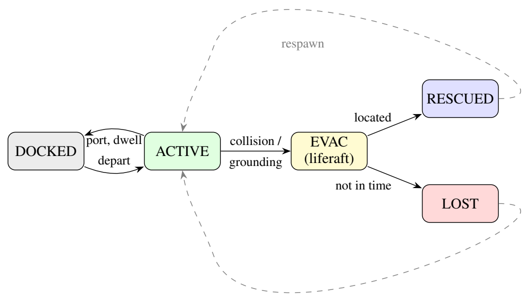

Figure 3: *Vessel state machine. A vessel cycles* ACTIVE ↔ DOCKED *along its route; a foundering collision or grounding sends it to* EVAC *(liferaft), from which the SAR chain yields* RESCUED *(search detection, Eq. 21) or* LOST *(the survivors are not located before the liferaft survival window elapses). Terminal vessels respawn to maintain fleet size.*

the leg network from the hand-authored routes, derives per-leg flows from the fleet and traversal times, and sums the head-on, overtaking, and crossing contributions into a per-year N_c. For the present network this yields $N_c \approx 0.01$ collisions/yr, within the regulator acceptance of <1 collision/yr/fairway; comparing N_c between Baseline A and the Proposed System (whose weather-aware slow-down lowers traffic flow) provides an externally interpretable, regulator-aligned check that the fatigue-coupling mechanism produces frequency reductions of the expected order.

8. Search-and-Rescue Subsystem

When a collision founders a vessel (Eq. 15, $S_i < S^{\dagger}$) or a grounding forces an evacuation, the survivors enter a liferaft and the search-and-rescue (SAR) chain is triggered. A rescue agent is dispatched from the nearest shore base (a port) after a mobilisation delay $\Delta t_{\text{mob}} = 2$ ticks — a helicopter (80 kn) when the datum lies beyond 40 nm offshore, otherwise a patrol vessel (25 kn) [22]. The asset then progresses through mobilise, transit, search, and embark phases; the subsystem layers two accredited components onto this transit. A second, parallel SAR chain handles the persons cast overboard from struck vessels that stay afloat (Eq. 16); it shares the same drift and random-search detection model but resolves each casualty individually (Section 8.3). Figure 3 shows the resulting vessel state machine.

8.1. Drift and Random-Search Detection (MASSIM)

Because the exact distress position is uncertain and the liferaft drifts, detection is modelled with the Koopman random-search formulation used in the MASSIM maritime-search model [23, 24]. A searcher of sweep width w moving at speed v_s over a search area A has instantaneous detection rate $\gamma(t) = w\,v_s/A$. The cumulative detection probability for n searchers over elapsed search time τ is

$$\text{CDP} = 1 - \exp\Big(-\frac{n\,w\,v_s\,\tau}{A}\Big) = 1 - e^{-C}, \qquad (21)$$

where $C = n\,w\,v_s\,\tau/A$ is the search coverage factor and v_s the asset's (season-degraded, Section 8.2) on-scene speed. The search area expands as the datum drifts: with an initial uncertainty radius r_0 and drift speed u_d, the area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$, so detection becomes harder the longer the response is delayed — the quantitative link from response latency to survival. Each on-scene tick advects the liferaft datum along the prevailing current and draws a Bernoulli detection at the per-tick coverage $1 - e^{-\mathrm{d}C}$ [23].

8.2. Environmental Degradation (Ashrafi)

Rescue-asset performance degrades with season following the Arctic SAR framework of Ashrafi et al. [25], in which air temperature, wind speed, significant wave height, and a wind-chill index combine into a per-month severity. The wind-chill index uses the standard form

$$\text{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}, \qquad (22)$$

with air temperature T (°C) and wind speed U (km/h). Rather than re-deriving the expert-assigned metocean thresholds, we adopt the framework's published month-group aggregation: each calendar month maps to one of four severity groups — A (May–August, mildest), B (April, September), C (March, October, November), and D (January, February, December, harshest) [25]. The group multiplicatively reduces the asset's transit and search speed — helicopter $1.00/0.95/0.77/0.60$ and patrol vessel $1.00/0.80/0.40/0.21$ for groups A–D, relative to the mildest group — and sets the per-survivor boarding time (2.1–10.7 min by helicopter, 6.1–31.2 min by vessel) [25]. On location, the asset embarks the n_b survivors at this season-degraded rate before the case is logged RESCUED. Transit, search (Eq. 21), and boarding therefore all lengthen in winter, so — reproducing the source case study — the longest rescue times occur December–February and the shortest May–August [25].

8.3. Man-Overboard Search

The casualties cast into the water by a struck-but-afloat vessel (Eq. 16) are handled as an explicit person-in-water search (Fig. 4), following the man-overboard simulation of Karatas et al. [23]. Each collision spawns an *incident*: its $\mathbb{E}[\text{overboard}]$ casualties become individual persons-in-water, scattered about the collision point and each assigned an independent drift bearing, so a search may recover some of an incident's people before others. One searcher — selected and dispatched by the same nearest-base, helicopter-versus-patrol rule as the liferaft chain — is tasked per incident and homes on the cluster centroid. While on scene it draws the Koopman per-tick detection of Eq. 21 *independently for every person*, recovering each as it is located. A person not recovered before the cold-water survival window $\tau_{\text{mob}} = 12$ ticks (≈ 3 h) elapses is lost; this is the model's only path to a crew fatality. The window is far shorter than the 96-tick liferaft endurance, reflecting unprotected immersion, and shrinks the effective margin in winter through the same Ashrafi speed degradation that slows the searcher (Section 8.2). The two SAR chains thus differ only in granularity: the liferaft recovers a vessel's crew as one whole-group case, the man-overboard search resolves individuals.

9. Experimental Design

9.1. Methods Under Test

The three conditions of Section 2 are realised as configuration switches over one agent codebase:

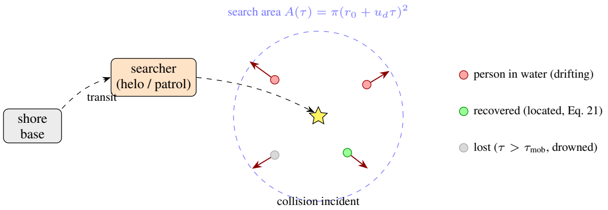

Figure 4: *Man-overboard search. A collision spawns one* incident *whose casualties (Eq. 16) become individual persons-in-water, each scattered about the impact point and advected along its own drift bearing (red arrows). A single searcher — chosen by the same nearest-base, helicopter-versus-patrol rule as the liferaft chain — transits from a shore base and homes on the cluster centroid, then draws the Koopman detection (Eq. 21)* independently per person *over the drift-growing search area, so individuals are recovered (green) as they are located. Anyone not reached before the cold-water survival window τ_{mob} elapses is lost (grey). Unlike the whole-group liferaft case of Fig. 3, each casualty resolves separately.*

- **Baseline A (Classical):** weather-unaware; nominal fair-weather VHF but the same environmental storm-comms degradation as every method (no privileged radio); no weather-induced slowdown; unmanaged fatigue ($m = 1.00$); collision-consequence multiplier highest.
- **Baseline B (Shore Rest Alerts):** weather-aware slowdown (Section 10.3); reduced fatigue build-rate ($m = 0.82$); weather/storm-gated comms.
- **Proposed System:** smart watch scheduling ($m = 0.65$) plus all Baseline B mitigations.

9.2. Scenarios

Table 2 summarises the four scenarios; each runs 30 independent seeds for 2880 ticks (30 days). With three methods and four scenarios, the full plan executes $4 \times 3 \times 30 = 360$ runs, dispatched in parallel.

9.3. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours accumulate 0.25 h per ACTIVE vessel per tick. A *fatality* is now an emergent outcome rather than an instantaneous draw: it is a person cast overboard (Eq. 16) or a liferaft occupant who is not recovered before their survival window elapses, so `fatal_per_1k_hrs` and `survival_ratio` measure the joint performance of the consequence and search-and-rescue chains.

Alongside the six hypothesis KPIs, the engine accumulates *diagnostic* telemetry counters that characterise the incident mix without entering the statistical battery: the collision-type breakdown (head-on, front-to-side, side-to-side counts from Eq. 17) and the man-overboard tallies (persons overboard, recovered, and lost). These are surfaced live on the dashboard and in the per-run snapshot to explain *why* a given fatality or survival figure arose — for instance, distinguishing a benign run dominated by low-energy overtaking contacts from one with foundering head-on impacts.

Crew preparedness (KPI 6) is accumulated each tick for every ACTIVE vessel as the product of a weather-readiness and a fatigue-readiness term, with method awareness factor

Table 2: *Scenario summary. All scenarios use 30 seeds and 2880 ticks/run (30 days at 15 min/tick).*

Scenario	Key conditions	Ships	Hyp.
1 — Calm Passage	calm-biased weather; nominal comms. Calibration target <0.5 collisions / 1 k ship-hrs.	20	Calib.
2 — Storm Corridor	moving storm zone; comms rate as low as 0.08 inside.	25	H1, H3
3 — Blind Shore	degraded coordination away from dense lanes: fleet-wide comms reliability 0.55 and a reduced avoidance berth.	25	H1
4 — Deep Water Rescue	sparse fleet far offshore: slowed SAR assets (55/16 kn) and longer mobilisation lengthen every rescue transit, stressing the SAR chain.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities $\div$ ship-hrs $\times 10^3$	H1
2	`collision_per_1k_hrs`	Collisions $\div$ ship-hrs $\times 10^3$	H1
3	`evac_activation_rate`	Evac activations $\div$ ship-hrs $\times 10^3$	H1
4	`survival_ratio`	$1-$ fatalities / crew exposed in collisions	H2
5	`avg_tta_hours`	Mean rescue Time-to-Arrival across dispatches	H2
6	`mean_p_prep`	Time-averaged crew preparedness across vessels	H1, H3

$\alpha_M \in \{2.0\,(\mathrm{A}),\ 1.3\,(\mathrm{B}),\ 1.0\,(\mathrm{Proposed})\}$:

$$P_{\text{prep}} = \max(0,\ 1 - \alpha_M W) \cdot \max(0,\ 1 - 0.70\,F). \quad (23)$$

9.4. Statistical Analysis

For every (KPI $\times$ scenario $\times$ method-pair), the two methods are compared with a two-tailed Mann–Whitney U test (average-rank tie handling, normal approximation): with $\mu_U = n_1 n_2/2$, $\sigma_U = \sqrt{n_1 n_2(n_1 + n_2 + 1)/12}$, $z = (U - \mu_U)/\sigma_U$, and $p = 2\Phi(-|z|)$, where Φ is evaluated by the Abramowitz–Stegun 26.2.17 rational approximation. The effect size is Cliff's $\delta = (U_1 - U_2)/(n_1 n_2) \in [-1, 1]$ [26]. Multiplicity is controlled by Holm–Bonferroni step-down over all tests; a result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if additionally $|\delta| \geq 0.2$. Method pairs tested are (A vs Proposed), (B vs Proposed), and (A vs B). The same seed sequence is applied to all conditions, eliminating seed-selection bias from pairwise comparisons.

10. Implementation

10.1. Engine and Execution

The model is implemented in Rust on the krABMaga agent-based framework [9], with an axum/tokio REST + WebSocket server for live run control and a rayon-parallel batch runner; a React/Leaflet frontend renders vessels, ports, the storm zone, the weather grid, the comms log, and KPIs live. Batch runs write per-run JSONL tick logs, a manifest, and a `summary.csv` for analysis. Vessel agents are scheduled at krABMaga ordering 200 and a single post-tick agent at ordering 1000.

10.2. Simulation Loop

Each macro tick runs four ordered phases (Fig. 5) whose ordering is a causal constraint: (1) *weather* — advance the field and apply any scenario-scripted event; (2) *perception & communication* — each vessel samples local hazard, updates fatigue (Eqs. 5–8), and CPA-screened encounters attempt their warning exchange (Eq. 11); (3) *agent step* — vessels navigate at hazard-gated speed (Section 10.3) or hold for avoidance, and rescue agents decrement mobilisation or advance under the SAR penalty (Section 8); (4) *physics & housekeeping* — hard-collision and grounding checks, SIMCOL consequence evaluation (Section 6), terminal-vessel respawn, logging, and clock advance.

10.3. Speed Reduction under Hazard

Two multiplicative slow-downs are applied by interpolating the just-moved position back toward the last valid position by a factor f. The hazard-gated factor (Baseline B and Proposed only; Baseline A ignores weather) is

$$f_W = \max\left(0.30,\ 1 - 0.45\,W\right), \quad (24)$$

and a storm-zone factor $f_{\text{storm}} \in \{1.0\,(\mathrm{A}),\ 0.45\,(\text{storm corridor}),\ 0.70\,(\text{default})\}$ damps speed inside the moving zone, acting as an implicit braking mechanism that lowers closing rates in deteriorating weather.

10.4. Weather Process Parameters

The seven-stage field of Section 3.5 uses: regime flip probabilities $p_{cs} = 0.03$, $p_{sc} = 0.08$ (Mixed preset;

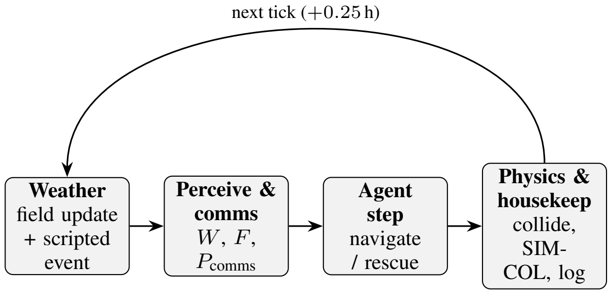

Figure 5: *The four ordered phases of each* 15*-minute macro tick. Data flows strictly left to right within a tick; weather precedes perception so a vessel acts on the field issued in the same tick, and physics/housekeeping closes the tick before the clock advances.*

scaled $\times(0.35, 1.40)$ Calm, $\times(2.00, 0.50)$ Stormy); reversion targets Calm$(0.30, 0.80, 0.25)$ / Storm$(0.72, 0.30, 0.75)$; up to $K_{\max} = 8$ storm cells with intensity decay $\tau \in \{0.006\,\text{storm}, 0.020\,\text{calm}\}$, radius $r = 8(1 + 0.15\sin\theta)\sqrt{I}$, and removal at $I < 0.12$; an AR(1) background

$$X_{t+1} = \rho_X\,\mathrm{adv}(X_t) + (1-\rho_X)\,\mu_X + \sigma_X(0.3 + \mathfrak{u})\,\eta_t, \quad (25)$$

with $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$, unpredictability $\mathfrak{u} = 0.30$, and $\eta \sim \mathcal{U}(-1, 1)$; a system overlay $\omega = (1 - \sqrt{d^2/r^2})^2 I$ adding to s, u and subtracting 0.85ω from v; cross-channel coupling $s \mathrel{+}= 0.35(u - 0.5) - 0.20(v - 0.5)$, $u \mathrel{-}= 0.25(v - 0.5)$; and a 5-point smoother with gradient cap $G_{\max} = 0.12$.

11. Discussion

The model's defensible core is the explicit, quantified coupling in Eq. 11: a single collision-warning exchange succeeds only when both the environment and both crews permit it. This makes the central hypotheses (H1–H3) mechanistic rather than correlational — fatigue management (m in Eq. 7) acts on outcomes solely by changing how often warnings are delivered and how prepared crews are, with no other lever distinguishing the conditions. Re-baselining the report on the real Baltic–North Sea engine, with hand-authored tracks and manual ports, keeps the experiment controlled while remaining geographically realistic.

The accredited layers raise the model from an internal A/B comparison toward external validity. SIMCOL (Section 6) grounds casualty counts in collision mechanics rather than a tunable rate; IWRAP (Section 7) places the resulting frequencies on a regulator-recognised scale; and the SAR subsystem (Section 8) makes `survival_ratio` and `avg_tta_hours` meaningful by modelling drift, random-search detection, and environmental degradation rather than instantaneous rescue. The SIMCOL external dynamics are pinned to the Pedersen–Zhang closed form, with the absorbed energy carried by the normal closing-velocity component and Lützen's added-mass coefficients (Eq. 14); the only modelling surrogate in the consequence chain is the survival factor S_i (Eq. 15), shaped to the SOLAS II-1 s-factor because the engine does not resolve per-ship residual stability. The SAR environmental degradation is likewise pinned to Ashrafi's published month-group constants (Section 8.2). The optional force-field repulsion exponent (Section 5.4) is the one constant its source does not fix—it is set there by coarse optimisation with AIS calibration still in development—so rather than hardcode a false-precision value the model exposes it as a tunable parameter defaulting to the standard inverse-square form.

Limitations follow from the controlled design. Hand-authored tracks trade empirical traffic realism for reproducibility; the deterministic lower-id give-way rule is a simplification of full COLREGs assignment; and the optional force-field track (Section 5.4) is not the default. Calibration targets — notably fewer than 0.5 collisions per $1\,000$ ship-hours under Baseline A in the Calm Passage scenario — are grounded in AIS-derived near-miss encounter statistics [6, 7].

12. References

[1] International Maritime Organization, "Human element – addressing the causes of accidents and incidents at sea," International Maritime Organization, London, UK, IMO Circular MSC-MEPC.7/Circ.7, 2018. [Online]. Available: https://wwwcdn.imo.org/localresources/en/OurWork/HumanElement/Documents/MSC-MEPC.7-Circ.7.pdf

[2] European Maritime Safety Agency, "Annual overview of marine casualties and incidents 2023," EMSA, Lisbon, Portugal, Tech. Rep., 2023. [Online]. Available: https://www.emsa.europa.eu/accident-investigation-publications/annual-overview.html

[3] International Union of Marine Insurance, "IUMI ocean hull statistics 2023," IUMI, Hamburg, Germany, Tech. Rep., 2023. [Online]. Available: https://iumi.com/statistics

[4] International Maritime Organization, "Convention on the international regulations for preventing collisions at sea, 1972 (COLREGs)," International Maritime Organization, London, UK, Tech. Rep., 1972. [Online]. Available: https://www.imo.org/en/About/Conventions/Pages/COLREG.aspx

[5] C.-k. Tam, R. Bucknall, and A. Greig, "Review of collision avoidance and path planning methods for ships in close range encounters," *Journal of Navigation*, vol. 62, no. 3, pp. 455–476, 2010.

[6] P. A. M. Silveira, A. P. Teixeira, and C. G. Soares, "Use of AIS data to characterise marine traffic patterns and ship collision risk off the coast of portugal," *Journal of Navigation*, vol. 66, no. 6, pp. 879–898, 2013.

[7] W. Zhang, F. Goerlandt, P. Kujala, and Y. Wang, "An advanced method for detecting possible near miss ship collisions from AIS data," *Ocean Engineering*, vol. 107, pp. 60–69, 2015.

[8] C. Chauvin, S. Lardjane, G. Morel, J.-P. Clostermann, and B. Langard, "Human and organisational factors in maritime accidents: Analysis of collisions at sea using the HFACS," *Accident Analysis & Prevention*, vol. 59, pp. 26–37, 2013.

[9] A. Antelmi, G. Cordasco, D. D'Ambrosio, D. De Vinco, and C. Spagnuolo, "Reliable and efficient agent-based modeling with krABMaga," *Journal of Artificial Societies and Social Simulation*, vol. 27, no. 2, p. 4, 2024.

[10] EMODnet Bathymetry Consortium, "EMODnet digital bathymetry (DTM 2024)," European Marine Observation and Data Network (EMODnet), Bathymetry portal, 2024, gridded water-depth product at $1/16'$ ($\approx$115 m) resolution; retrieved via the OGC WCS endpoint `ows.emodnet-bathymetry.eu/wcs`. [Online]. Available: https://emodnet.ec.europa.eu/en/bathymetry

[11] EMODnet Human Activities, "Vessel density map," European Marine Observation and Data Network (EMODnet), Human Activities portal, 2023, AIS-derived ship density (hours $\mathrm{km}^{-2}\,\mathrm{month}^{-1}$) on a 1 km grid, resolved by vessel type; retrieved via the OGC WCS endpoint `ows.emodnet-humanactivities.eu/wcs`. [Online]. Available: https://emodnet.ec.europa.eu/en/human-activities

[12] PIANC, "Harbour approach channels design guidelines (MarCom Report no. 121)," World Association for Waterborne Transport Infrastructure (PIANC), Brussels, Belgium, Tech. Rep., 2014.

[13] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[14] F. Xiao, H. Ligteringen, C. van Gulijk, and B. Ale, "Nautical traffic simulation with multi-agent system," in *Proceedings of the 16th International IEEE Annual Conference on Intelligent Transportation Systems (ITSC)*. The Hague, The Netherlands: IEEE, 2013, pp. 1245–1252.

[15] M. Lützen, "Ship collision damage," Ph.D. dissertation, Technical University of Denmark, Department of Mechanical Engineering, Kongens Lyngby, Denmark, 2001.

[16] P. T. Pedersen and S. Zhang, "On impact mechanics in ship collisions," *Marine Structures*, vol. 11, no. 10, pp. 429–449, 1998.

[17] P. T. Pedersen, "Collision and grounding mechanics," *Proceedings of the Danish Society of Naval Architecture and Marine Engineering (WEMT)*, pp. 125–157, 1995.

[18] V. U. Minorsky, "An analysis of ship collisions with reference to protection of nuclear power plants," *Journal of Ship Research*, vol. 3, no. 2, pp. 1–4, 1959.

[19] International Maritime Organization, "SOLAS chapter II-1: Construction — subdivision and stability (probabilistic damage stability)," International Maritime Organization, London, UK, International Convention for the Safety of Life at Sea, 2020.

[20] P. Friis-Hansen, "Basic modelling principles for prediction of collision and grounding frequencies (IWRAP Mk II working document)," Technical University of Denmark, Kongens Lyngby, Denmark, Tech. Rep., 2008, rev. 4.

[21] Z. Lusic and M. Ćorić, "Models for estimating the potential number of ship collisions," *Journal of Navigation*, vol. 68, no. 4, pp. 735–749, 2015.

[22] International Maritime Organization, "International aeronautical and maritime search and rescue (IAMSAR) manual," IMO / ICAO, London, UK, Tech. Rep., 2019.

[23] M. Karatas, B. S. Onggo, and M. M. Gunal, "Using simulation to measure the performance of a search operation for a man overboard," in *Proceedings of the 2018 Winter Simulation Conference (WSC)*. Gothenburg, Sweden: IEEE, 2018, pp. 2742–2753.

[24] B. O. Koopman, *Search and Screening: General Principles with Historical Applications*. New York: Pergamon Press, 1980.

[25] B. Ashrafi, G. Kim, M. Naseri, J. Barabady, S. Dhar, G. Heo, and S. Baek, "An agent-based modelling framework for performance assessment of search and rescue operations in the Barents Sea," *Safety in Extreme Environments*, vol. 6, pp. 183–200, 2024.

[26] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychological Bulletin*, vol. 114, no. 3, pp. 494–509, 1993.

A. Default Constants Reference

Table 4 lists the principal engine constants with default values; all are overridable via `SimulationConfig` or the dashboard.

Table 4: *Selected engine default constants.*

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	$(1.0, 0.5, 0.8)$	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuvre lockout
r_{coll}	0.05 nm	hit at $2r = 0.10$ nm
port collision exclusion	1 nm	VTS harbour water
follow vicinity / gap	3 / 0.5 nm	speed-match / keep-distance
same-destination	5 nm	endpoint match radius
$(m_{\text{sway}}, m_{\text{surge}})$	$0.85, 0.05$	SIMCOL added mass
Minorsky a, b	$47.2, 32.7$	energy–volume (MJ, m^3)
$\ell/L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^\dagger$	$0.10, 0.30, 0.50$	S_i knee / loss / founder
$(\beta_{\text{base}}, \beta_W, \beta_c)$	$0.0015, 0.0040, 0.0008$	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^*, κ	0.40, 0.65	comms-degradation knee
storm comms rate	0.08	corridor link reliability
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
helo / patrol cutoff	40 nm	SAR asset-selection datum range